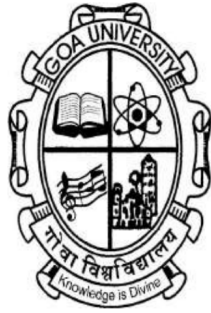


GOA UNIVERSITY



FPGA-Accelerated Neuromorphic Keyword Spotting: A Cochlea-to-SNN Pipeline

*A project report submitted in partial fulfillment of the requirements
for the award of the degree of*

Bachelor of Engineering in
VLSI Design & Technology

by

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This work, conducted under my supervision during the academic year 2025–2026, is a genuine endeavor. It was undertaken as a partial fulfillment of the prerequisites for obtaining the Bachelor of Engineering degree in VLSI Design & Technology at Goa College of Engineering, Farmagudi, Ponda Goa, affiliated with Goa University.

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Abstract

This project focuses on developing a real-time neuromorphic speech classification system implemented on a Field-Programmable Gate Array (FPGA). The core objective is to emulate the human auditory pathway, creating an ultra-low-power system for keyword detection. The processing begins by converting incoming audio signals into biologically plausible spike trains using a computational model of the cochlea. This mimics the initial stages of sound processing in the inner ear. These generated spike trains are then fed into a Spiking Neural Network (SNN) deployed directly onto the FPGA hardware. The SNN, inspired by the brain's neural circuitry, processes the temporal information from the spikes. It utilizes neuron models like the Leaky Integrate-and-Fire (LIF) model to simulate neural dynamics and can employ learning rules such as Surrogate Gradient Descent to adapt its synaptic weights, enhancing its recognition accuracy over time. By leveraging the massive parallelism and energy efficiency of FPGAs for SNN execution, this system achieves real-time performance while consuming minimal power. The final outcome is a hardware-efficient system capable of interpreting spoken commands by translating acoustic signals into neural-like activity and processing them through a brain-inspired computational framework, paving the way for always-on intelligent auditory interfaces in edge devices.

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1 Introduction

1.1 Background

The rapid growth of intelligent electronic systems has increased the demand for efficient and reliable speech-based human-machine interaction. Voice-controlled assistants, smart home devices, wearable electronics, and edge-AI platforms increasingly rely on continuous auditory monitoring and keyword detection to provide intuitive user interfaces. As the number of such devices grows, the need for speech-processing systems that can operate in real time while consuming minimal power has become increasingly important.

Conventional speech recognition systems typically employ deep learning algorithms executed on high-performance processors or graphics processing units (GPUs). Although these approaches achieve high recognition accuracy, they often require significant computational resources, memory bandwidth, and power consumption. Such requirements pose challenges for deployment in embedded and battery-powered systems where energy efficiency and low latency are critical design constraints.

In contrast, biological auditory systems demonstrate remarkable efficiency. The human auditory pathway is capable of processing complex acoustic environments, distinguishing speech from background noise, and recognising sounds with exceptional robustness while consuming only a small amount of energy. This efficiency arises from highly specialised neural structures that process information in an event-driven manner rather than through continuous high-rate numerical computation.

Inspired by these biological principles, neuromorphic engineering seeks to develop hardware systems that emulate the structure and operation of neural circuits. Neuromorphic systems represent information using discrete spike events and perform computation only when relevant activity occurs, thereby reducing unnecessary processing and energy consumption. This event-driven paradigm has attracted considerable interest for sensory processing applications, particularly in vision and audition.

Auditory neuromorphic systems are designed to replicate the signal-processing stages of the biological hearing pathway. Sound is first decomposed into frequency-selective channels analogous to the basilar membrane of the cochlea, transformed into neural activity through inner hair cell models, and subsequently represented as spike trains similar to those

transmitted by auditory nerve fibres. These spike-based representations can then be processed by Spiking Neural Networks (SNNs), which offer a biologically inspired alternative to conventional artificial neural networks.

Field-Programmable Gate Arrays (FPGAs) provide a suitable platform for implementing such systems due to their inherent parallelism, reconfigurability, and ability to perform real-time signal processing. By combining biologically inspired auditory models with FPGA hardware, it becomes possible to develop efficient neuromorphic speech-processing systems suitable for edge-AI and always-on listening applications.

This project explores the implementation of an FPGA-based neuromorphic auditory processing pipeline that converts audio signals into spike-based representations for subsequent keyword-spotting and speech-classification tasks.

1.2 Motivation

The motivation for this project arises from the convergence of several technological requirements and research trends in the fields of speech processing, neuromorphic engineering, and embedded hardware design.

One of the primary motivations is the growing demand for energy-efficient speech-processing systems at the edge. Modern voice-enabled devices such as smart assistants, hearing aids, wearable electronics, environmental monitoring systems, and Internet of Things (IoT) platforms increasingly require continuous auditory monitoring while operating under strict power constraints. Conventional speech-recognition pipelines typically rely on computationally intensive feature extraction and machine learning algorithms executed on general-purpose processors or dedicated digital signal processors. Such approaches can consume significant computational resources and energy, making them less suitable for always-on embedded applications. A neuromorphic auditory front-end offers a promising alternative by exploiting sparse, event-driven processing, where computation occurs only when meaningful signal activity is present. This approach has the potential to significantly reduce power consumption while preserving essential auditory information.

Another important motivation is the remarkable efficiency of the biological auditory system. The human cochlea performs sophisticated frequency analysis, dynamic range compression, and neural encoding using highly parallel biological mechanisms. These processes transform acoustic signals into sequences of neural spikes that efficiently represent auditory information. Emulating these biological principles in hardware provides an opportunity to develop compact and efficient auditory processing systems that retain the temporal characteristics of speech while reducing computational complexity. The study of cochlear models, inner hair cell transduction mechanisms, and spike-based neural representations therefore offers valuable insights into alternative approaches to speech processing.

The increasing interest in Spiking Neural Networks (SNNs) further strengthens the relevance of neuromorphic auditory systems. Unlike conventional artificial neural networks, SNNs process information through discrete spike events and naturally operate on temporal data. Since neuromorphic auditory front-ends produce spike-based representations of sound, they provide a suitable interface for SNN-based classification systems. This compatibility enables the development of end-to-end neuromorphic pipelines capable

of performing speech recognition and keyword spotting using biologically inspired computational principles.

Another motivation for this project is to explore the implementation of biologically inspired auditory processing algorithms directly in digital hardware. FPGA platforms provide a flexible environment for translating mathematical and neuroscience-based models into synthesizable hardware architectures. By implementing the cochlear filterbank, inner hair cell model, spike generation mechanisms, and Address Event Representation (AER) communication framework in Verilog HDL, the project investigates the practical challenges associated with fixed-point arithmetic, resource utilization, timing constraints, and hardware verification.

Furthermore, neuromorphic systems are inherently designed to exploit sparse and event-driven computation, where processing occurs only when relevant neural activity is present. This characteristic has the potential to reduce unnecessary computations and improve energy efficiency compared to conventional continuously operating signal-processing pipelines. Investigating such architectures on FPGA hardware provides valuable insight into the development of future low-power auditory processing systems for edge-AI applications.

Although the current work focuses primarily on the development and validation of the neuromorphic auditory processing pipeline, the resulting architecture forms a foundation for future integration with spiking neural network classifiers and real-time speech-processing systems.

Finally, the project offers significant educational and research value. The implementation of a neuromorphic auditory processing pipeline requires the integration of concepts from digital signal processing, neuroscience, machine learning, and digital hardware design. Designing such a system from mathematical modelling through hardware implementation and verification provides a comprehensive understanding of modern VLSI system design methodologies while exposing the practical challenges involved in translating biologically inspired algorithms into FPGA-based hardware systems. The project therefore serves as both a learning platform and a stepping stone toward future research in neuromorphic computing, intelligent edge devices, and low-power sensory processing systems.

1.3 Biological Basis

Before describing the hardware implementation, it is important to understand the biological auditory system that inspires the proposed neuromorphic architecture. The human auditory pathway converts acoustic pressure waves into electrical impulses that are transmitted to the brain for interpretation. This project models the early stages of this pathway, namely the cochlea, the inner hair cells, and the auditory nerve fibres. These biological structures perform frequency analysis, signal transduction, and neural encoding, respectively, forming the basis of the FPGA-based neuromorphic auditory processing pipeline developed in this work.

1.3.1 The Human Cochlea

The cochlea is a spiral-shaped organ located within the inner ear and is responsible for transforming incoming sound waves into frequency-selective mechanical vibrations. When sound enters the ear, vibrations transmitted through the middle ear create pressure waves within the cochlear fluid, causing the basilar membrane to vibrate.

The basilar membrane exhibits a gradual variation in its mechanical properties along its length. It is narrow and stiff near the base and wider and more flexible near the apex. As a result, different locations along the membrane respond preferentially to different frequencies. High-frequency sounds produce maximum displacement near the base, while low-frequency sounds produce maximum displacement near the apex. This spatial frequency mapping is known as tonotopic organisation and forms the foundation of biological auditory frequency analysis.

Unlike a conventional Fourier transform, the cochlea performs frequency decomposition through a bank of overlapping filters whose bandwidths vary with frequency. The frequency selectivity of these filters is commonly described using the Equivalent Rectangular Bandwidth (ERB) scale, which approximates the bandwidth of auditory filters across the human hearing range.

Since ERB bandwidth increases with frequency, auditory filters become progressively wider at higher frequencies. Consequently, more filters are allocated to lower frequencies, where speech contains significant perceptual information. To emulate this behaviour, the proposed system employs a 16-channel ERB-spaced cochlear filterbank that performs frequency decomposition of the incoming audio signal.

Another important characteristic of the cochlea is its nonlinear response

to sound intensity. Biological hearing compresses a wide dynamic range of acoustic signals into a smaller neural operating range. This phenomenon motivates the nonlinear processing stages incorporated later in the auditory pipeline.

1.3.2 Inner Hair Cells

Located within the organ of Corti are specialised sensory receptors known as Inner Hair Cells (IHCs). These cells convert the mechanical vibrations of the basilar membrane into electrical signals that can be processed by the nervous system.

When the basilar membrane vibrates, the stereocilia of the inner hair cells are deflected, producing changes in the cell membrane potential. The resulting electrical response contains several important signal-processing characteristics that influence auditory perception.

The first characteristic is rectification. Inner hair cells respond more strongly to motion in one direction than the other, causing the output signal to resemble a half-wave-rectified version of the basilar membrane motion. This biological behaviour forms the basis of the half-wave rectification stage used in the hardware model.

A second characteristic is adaptation. Following a sustained stimulus, the response of the inner hair cell gradually decreases with time. Adaptation enhances sensitivity to signal onsets and temporal changes while reducing sensitivity to constant background signals. This behaviour is modelled using an adaptation filter in the proposed architecture.

A third characteristic is envelope extraction. Due to the low-pass nature of the inner hair cell membrane, rapid oscillations of the input signal are smoothed, preserving the slower-varying amplitude envelope. Since speech information is strongly represented within these envelopes, envelope extraction plays an important role in auditory processing.

To emulate these biological functions, the inner hair cell model implemented in this project consists of nonlinear compression, half-wave rectification, adaptation filtering, and envelope extraction stages. Together, these blocks transform the filterbank outputs into biologically plausible neural excitation signals suitable for spike generation.

1.3.3 Auditory Nerve Fibres

The outputs of the inner hair cells are transmitted to the brain through auditory nerve fibres. Each fibre responds preferentially to a specific frequency region and conveys information using sequences of discrete electrical impulses known as action potentials or spikes.

One of the most important coding mechanisms employed by auditory nerve fibres is rate coding. As stimulus intensity increases, the average firing rate of the corresponding nerve fibre also increases. Thus, stronger acoustic stimuli produce higher spike rates, while weaker stimuli generate fewer spikes.

Auditory nerve fibres also exhibit a refractory period following each spike, during which the neuron is temporarily unable to generate another action potential. This property limits the maximum firing rate and contributes to the temporal structure of neural activity.

Another important feature is tonotopic organisation. Since each auditory nerve fibre is connected to a specific location along the basilar membrane, it inherits a characteristic frequency corresponding to that cochlear region. Consequently, auditory information is transmitted as a collection of parallel spike streams, each representing activity within a particular frequency band.

In the proposed system, auditory nerve behaviour is modelled using a bank of Leaky Integrate-and-Fire (LIF) neurons. The membrane potential of each neuron integrates the processed inner hair cell output and generates spikes whenever a predefined threshold is exceeded. Stronger inputs result in more frequent spikes, thereby implementing biological rate coding. A refractory mechanism is incorporated to emulate the recovery behaviour observed in biological neurons.

The generated spikes are subsequently encoded using Address Event Representation (AER), allowing neural activity from multiple frequency channels to be transmitted efficiently to downstream neuromorphic processing stages such as Spiking Neural Networks (SNNs).

1.4 Neuromorphic Engineering and FPGA's

Neuromorphic engineering is an interdisciplinary field that seeks to develop electronic systems inspired by the structure and operating principles of biological nervous systems. Introduced by Carver Mead in the late 1980s, the field combines concepts from neuroscience, electronics, computer engineering, and signal processing to design hardware capable of performing brain-inspired computation. Unlike conventional computing systems, which process information sequentially using clock-driven operations, neuromorphic systems employ event-driven processing and distributed computation similar to that observed in biological neural networks.

One of the fundamental characteristics of neuromorphic systems is the use of spikes as the primary means of information representation and communication. Rather than continuously processing data, computation occurs only when meaningful events are generated. This event-driven approach reduces unnecessary activity and forms the basis for the energy efficiency often associated with biological neural systems.

Another important feature of neuromorphic computing is its ability to process temporal information naturally. Biological sensory systems encode information not only through signal magnitude but also through the timing and pattern of neural spikes. Consequently, neuromorphic architectures are particularly well suited for sensory applications such as vision, audition, and robotics, where temporal dynamics play a significant role.

Neuromorphic systems also exploit parallelism extensively. Biological neural networks consist of large numbers of neurons operating simultaneously and independently. Similarly, neuromorphic hardware architectures often employ multiple processing elements working in parallel, enabling efficient handling of complex sensory data streams.

Field-Programmable Gate Arrays (FPGAs) provide an attractive platform for implementing neuromorphic systems. Their reconfigurable architecture allows designers to develop custom digital circuits tailored to specific computational models. Unlike software implementations executed on general-purpose processors, FPGA-based systems can exploit hardware-level parallelism and dedicated arithmetic resources to perform multiple operations concurrently.

Another advantage of FPGAs is their flexibility. Neuromorphic models can be modified, tested, and refined without requiring fabrication of dedicated integrated circuits. This makes FPGAs particularly suitable for research, prototyping, and educational applications where architectural

exploration and iterative design are important. Furthermore, FPGA development tools provide extensive support for simulation, verification, and performance analysis, enabling efficient evaluation of complex neuromorphic architectures.

In this project, an FPGA platform is used to implement the complete neuromorphic auditory processing pipeline, including cochlear frequency decomposition, inner hair cell modelling, spike generation using Leaky Integrate-and-Fire (LIF) neurons, and Address Event Representation (AER) communication. The inherent parallelism and configurability of FPGA hardware make it a suitable platform for investigating biologically inspired auditory processing systems and their potential application in future low-power intelligent devices.

1.5 Applications

Neuromorphic auditory processing systems have potential applications in a variety of domains where efficient and intelligent audio analysis is required.

Speech Recognition Systems: Neuromorphic front-ends can be employed in keyword spotting and speech-recognition systems for smart devices, voice-controlled appliances, and Internet-of-Things (IoT) applications. By converting audio signals into sparse spike-based representations, these systems can reduce computational requirements while maintaining useful speech information for classification.

Hearing Assistance Devices: The signal-processing stages implemented in this project are inspired by the biological auditory pathway and are closely related to those used in hearing aids and cochlear implants. Such systems can benefit from biologically inspired frequency decomposition and envelope extraction techniques for improved auditory perception.

Environmental Sound Monitoring: Neuromorphic audio processing can be used for continuous monitoring of environmental sounds such as alarms, machine faults, animal calls, and other acoustic events. The event-driven nature of spike-based processing makes it suitable for long-duration monitoring applications.

Edge Artificial Intelligence: As edge devices increasingly require local intelligence without reliance on cloud computing, neuromorphic systems offer an attractive solution due to their efficient processing of temporal sensory data. Audio-based human-machine interaction systems can particularly benefit from such architectures.

Neuromorphic Computing Research: The FPGA-based platform developed in this project provides a flexible framework for studying biologically inspired auditory processing and serves as a foundation for future research involving Spiking Neural Networks and event-based machine learning systems.

1.6 Challenges

The implementation of a neuromorphic auditory processing pipeline on FPGA presents several technical challenges. These challenges influenced both the system design and the hardware architecture adopted in this project.

1.6.1 Fixed-Point Hardware Implementation

The conversion of signal-processing algorithms from floating-point mathematical models to fixed-point hardware representations introduces quantisation effects and numerical limitations. Appropriate word lengths must be selected to maintain accuracy while avoiding overflow and excessive hardware utilisation.

1.6.2 Sampling and Timing Management

The auditory processing stages operate on audio samples acquired at a specific sampling frequency, whereas the FPGA operates at a significantly higher clock frequency. Ensuring correct synchronization between these domains is essential for preserving the intended signal-processing behaviour.

1.6.3 Biological Model Approximation

The human auditory system exhibits highly complex nonlinear and adaptive behaviour. Implementing an exact biological model in digital hardware is impractical; therefore, simplified computational models must be employed while retaining the important functional characteristics of cochlear processing.

1.6.4 Event-Based Communication

The conversion of neural spike activity into Address Event Representation (AER) packets requires efficient event handling and arbitration mechanisms. Multiple spike events may occur simultaneously, making reliable event encoding and transmission an important design consideration.

1.7 Report Organisation

The remainder of this report is organised as follows:

Chapter 2 – Literature Survey This chapter reviews previous work related to neuromorphic auditory processing, FPGA-based cochlear models, Spiking Neural Networks, FPGA implementations of SNNs, and Address Event Representation systems. The chapter identifies existing approaches and highlights the research gap addressed by this project.

Chapter 3 – Problem Statement This chapter presents the problem definition, project objectives, scope, and design constraints.

Chapter 4 – System Design and Implementation This chapter describes the overall architecture of the proposed system and details the implementation of each processing stage, including audio preprocessing, cochlear frequency decomposition, inner hair cell modelling, Leaky Integrate-and-Fire neurons, Address Event Representation encoding, and FPGA implementation.

Chapter 5 – Results and Discussion This chapter presents simulation results, hardware verification results, resource utilisation statistics, and performance analysis of the developed system.

Chapter 6 – Conclusion and Future Work This chapter summarises the major contributions of the project, discusses its limitations, and outlines possible directions for future enhancement.

2 Literature Survey

2.1 FPGA-Based Neuromorphic Audio Processing

The human auditory system has inspired the development of neuromorphic auditory processors capable of transforming sound signals into spike-based representations. Such systems attempt to replicate the functionality of the biological cochlea by decomposing incoming audio into multiple frequency channels and generating neural-like responses suitable for subsequent processing by spiking neural networks.

One of the earliest contributions in this field was the Lyon-Mead silicon cochlea, which demonstrated that cochlear frequency analysis could be implemented using electronic circuits inspired by biological auditory mechanisms. Subsequent developments introduced more sophisticated cochlear models incorporating nonlinear compression, adaptation, and tonotopic frequency organisation. These systems established the foundation for neuromorphic auditory processing and demonstrated the potential of event-driven computation for sound analysis.

With the increasing availability of programmable hardware platforms, several researchers have investigated FPGA-based implementations of cochlear models. FPGAs provide an attractive platform due to their reconfigurability, parallel processing capability, and ability to perform real-time signal processing. Various studies have implemented gammatone filterbanks, ERB-spaced filterbanks, and auditory feature extraction systems on FPGA devices for speech and audio applications.

Most FPGA-based auditory processors focus primarily on frequency decomposition and auditory feature extraction. While these systems successfully emulate important aspects of cochlear processing, many stop at the feature extraction stage and do not generate spike-based outputs suitable for direct integration with neuromorphic classifiers.

The present work extends this concept by implementing a complete neuromorphic auditory front-end that performs cochlear frequency decomposition, inner hair cell modelling, spike generation, and Address Event Representation encoding within a unified FPGA framework.

2.2 Spiking Neural Networks

Spiking Neural Networks (SNNs) represent the third generation of artificial neural network models and are inspired by the manner in which biological neurons communicate through discrete electrical impulses known as spikes. Unlike conventional artificial neural networks that process continuous-valued activations, SNNs encode information in the timing and frequency of spike events.

The fundamental computational unit of an SNN is the neuron model. Several neuron models have been proposed in the literature, including the Hodgkin-Huxley model, the Izhikevich model, the Integrate-and-Fire model, and the Leaky Integrate-and-Fire (LIF) model. Among these, the LIF model is widely adopted because it provides a good balance between biological realism and computational simplicity.

Recent advances in training methodologies such as Spike-Timing-Dependent Plasticity (STDP) and surrogate-gradient learning have significantly improved the performance of SNNs on practical machine-learning tasks. Consequently, SNNs have found applications in speech recognition, computer vision, robotics, autonomous systems, and edge artificial intelligence.

One of the major advantages of SNNs is their event-driven nature. Computation occurs only when spikes are generated, reducing unnecessary processing and enabling energy-efficient operation. This characteristic makes SNNs particularly suitable for deployment in low-power embedded systems and neuromorphic hardware platforms.

For auditory processing applications, SNNs are especially attractive because biological auditory information is naturally represented as spike trains. Neuromorphic cochlear systems can therefore provide a direct interface between sensory input and spike-based neural computation.

2.3 FPGA Implementations of SNNs

The implementation of Spiking Neural Networks on FPGA platforms has attracted significant research interest due to the parallel architecture and reconfigurable nature of FPGA devices. Unlike software-based implementations, FPGA-based SNNs can exploit hardware parallelism to achieve high throughput and low latency while maintaining relatively low power consumption.

Several researchers have implemented Leaky Integrate-and-Fire neurons, synaptic networks, and learning mechanisms on FPGA devices. These implementations typically employ fixed-point arithmetic to reduce hard-

ware complexity and improve resource efficiency. Hardware optimisation techniques such as resource sharing, pipelining, and memory-efficient neuron storage are commonly used to enable large-scale neural networks on FPGA platforms.

FPGA-based SNN accelerators have demonstrated promising results in applications including image recognition, speech classification, robotics, and event-based sensory processing. However, many existing implementations rely on externally generated spike datasets rather than generating biologically inspired spike trains directly on hardware.

Furthermore, several studies focus exclusively on neural network acceleration while treating sensory preprocessing as a separate software task. This separation limits the development of complete neuromorphic systems capable of processing sensory information entirely within hardware.

The work presented in this project contributes toward bridging this gap by providing an FPGA-based auditory front-end capable of producing spike trains suitable for direct integration with SNN classifiers.

2.4 Address Event Representation

Address Event Representation (AER) is one of the most widely adopted communication protocols in neuromorphic systems. It provides an efficient mechanism for transmitting neural spike events between processing blocks while preserving temporal information.

In conventional sampled-data systems, information is transmitted continuously regardless of whether significant events occur. In contrast, AER adopts an event-driven communication strategy in which data are transmitted only when spikes are generated. Each spike is represented by an address identifying the source neuron together with timing information indicating when the event occurred.

The use of AER significantly reduces communication bandwidth requirements because only active channels generate traffic. This property is particularly advantageous in neuromorphic systems where neural activity is sparse and event-driven by nature.

AER has been employed extensively in silicon retinas, silicon cochleae, neuromorphic vision sensors, and large-scale neuromorphic computing platforms. Modern neuromorphic processors frequently use AER or related event-based protocols to exchange information between neuron populations.

For auditory processing systems, AER enables efficient transmission of

spike events generated by cochlear channels and auditory neurons. The event-driven nature of AER aligns naturally with the sparse activity patterns produced by neuromorphic sensory front-ends.

In this project, AER is used to encode spike events generated by the Leaky Integrate-and-Fire neuron bank. Each transmitted packet contains information identifying the active channel and the corresponding event timing, enabling seamless integration with downstream neuromorphic processing systems.

2.5 Research Gap Analysis

A review of the existing literature reveals substantial progress in the individual areas of cochlear modelling, spiking neural networks, FPGA-based neural implementations, and event-driven communication systems. However, these components are often investigated independently.

Many FPGA-based cochlear implementations focus primarily on frequency decomposition and auditory feature extraction without generating biologically inspired spike trains. Similarly, FPGA-based SNN implementations generally assume the availability of pre-generated spike inputs and do not address auditory front-end processing. Research involving Address Event Representation typically concentrates on communication architectures rather than complete auditory processing pipelines.

Consequently, there exists a need for an integrated FPGA-based neuromorphic auditory system capable of performing frequency decomposition, inner hair cell modelling, spike generation, and event-based encoding within a unified hardware framework.

The present project addresses this gap by developing an end-to-end neuromorphic auditory processing pipeline implemented entirely on FPGA hardware. The system transforms raw audio signals into Address Event Representation packets through biologically inspired cochlear processing stages, thereby providing a suitable front-end for future spike-based speech recognition systems.

3 Problem Statement

3.1 Problem Definition

Modern speech-processing systems rely heavily on conventional digital signal processing and machine-learning algorithms executed on general-purpose processors or graphics processing units. While these approaches achieve high recognition accuracy, they often require significant computational resources and power consumption, making them less suitable for resource-constrained embedded and edge devices.

In contrast, biological auditory systems perform complex sound analysis with remarkable efficiency by employing event-driven neural processing. The human cochlea decomposes incoming sound into frequency-selective channels, while inner hair cells and auditory nerve fibres convert acoustic information into neural spike trains that are processed by higher auditory centres.

Neuromorphic engineering seeks to replicate these biological principles in electronic systems. However, many existing FPGA-based auditory processing systems focus only on frequency decomposition or feature extraction, while many Spiking Neural Network implementations assume the availability of externally generated spike inputs. Consequently, there exists a gap between auditory signal acquisition and spike-based neural processing.

The problem addressed in this project is the design and implementation of a complete neuromorphic auditory front-end on FPGA capable of transforming raw audio signals into biologically inspired spike-based representations. The system must emulate the key stages of peripheral auditory processing, including cochlear frequency decomposition, inner hair cell transduction, auditory nerve spike generation, and event-based communication through Address Event Representation (AER).

The generated spike events should provide a suitable interface for future integration with Spiking Neural Network classifiers used for speech and audio recognition tasks.

3.2 Objectives

The primary objective of this project is to develop and verify an FPGA-based neuromorphic auditory processing pipeline inspired by the human auditory system.

The specific objectives are as follows:

1. To study the biological mechanisms of auditory signal processing, including cochlear frequency analysis, inner hair cell transduction, and auditory nerve spike generation.
2. To design and implement an audio preprocessing stage consisting of DC offset removal and pre-emphasis filtering.
3. To develop a multi-channel cochlear filterbank based on the Equivalent Rectangular Bandwidth (ERB) scale for frequency decomposition of speech signals.
4. To implement an Inner Hair Cell (IHC) model incorporating nonlinear compression, rectification, adaptation, and envelope extraction.
5. To design and implement a bank of Leaky Integrate-and-Fire (LIF) neurons for converting auditory features into spike trains.
6. To develop an Address Event Representation (AER) encoder for efficient event-based communication of spike information.
7. To implement the complete system in Verilog HDL and deploy it on an FPGA platform.
8. To verify the functionality of each processing stage through simulation and hardware testing.
9. To generate spike-based auditory representations suitable for future integration with Spiking Neural Network classifiers.

3.3 Scope and Constraints

Scope

The scope of this project is limited to the implementation of a neuromorphic auditory front-end capable of converting audio signals into spike-based event streams. The system includes audio preprocessing, cochlear

frequency decomposition, inner hair cell modelling, Leaky Integrate-and-Fire neuron modelling, and AER packet generation.

The design is implemented entirely in digital hardware using Verilog HDL and targeted for FPGA deployment. The generated spike events are intended for use by downstream neuromorphic processing systems such as Spiking Neural Networks.

The project focuses primarily on the auditory preprocessing stages of a neuromorphic speech-recognition pipeline and establishes the foundation for future hardware-based neuromorphic classification systems.

Constraints

The design is subject to several practical constraints:

1. FPGA resource limitations, including logic elements, DSP resources, and memory availability.
2. Fixed-point arithmetic implementation, which introduces quantisation effects and numerical precision limitations.
3. Simplification of biological auditory mechanisms to achieve hardware feasibility and manageable computational complexity.
4. Limited number of cochlear channels compared to the thousands of biological auditory channels present in the human cochlea.
5. Use of prerecorded audio signals for system verification rather than continuous real-time microphone acquisition.
6. The scope of the present work is restricted to the neuromorphic auditory front-end, while the development and training of the final Spiking Neural Network classifier remain outside the primary implementation scope.

These constraints were considered during the design process to achieve a balance between biological inspiration, hardware complexity, and practical FPGA implementation.

4 System Design and Implementation

4.1 Overall System Architecture

The proposed neuromorphic auditory processing system consists of multiple stages that emulate the early auditory pathway of the human hearing system. The input audio signal first undergoes preprocessing to remove unwanted components and enhance high-frequency information. The preprocessed signal is then passed through a cochlear filterbank for frequency decomposition. The outputs of the filterbank are processed by the Inner Hair Cell (IHC) model, followed by a bank of Leaky Integrate-and-Fire (LIF) neurons that convert the analog features into spike trains. Finally, the generated spikes are encoded using Address Event Representation (AER) for transmission to subsequent neuromorphic processing stages.

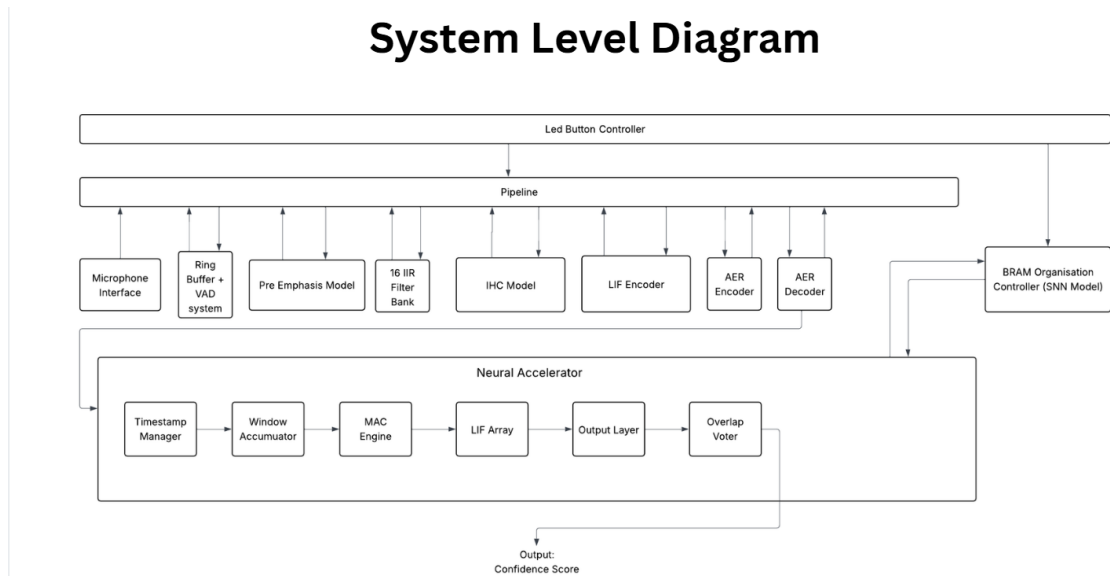


Figure 4.1: Overall architecture of the proposed neuromorphic auditory processing pipeline.